DIGITAL SIGNAL ENCODER AND DECODER

ITT 036 · Programming Assignment 1 · Autumn 2026

Student: Adnan Nazir **Enrollment:** 2024BITE065

Language: Java **Visualization:** Java Swing and AWT

1. Objective

Build a modular Java application that generates binary data, finds the longest palindromic substring, implements digital line encoding and decoding, and demonstrates analog-to-digital conversion using Pulse Code Modulation (PCM) and Delta Modulation (DM).

2. Main Features

- **Data generation and palindrome detection:** Creates random binary sequences, including sequences with four or eight consecutive zeros, and finds the longest palindrome.
- **Line encoding and decoding:** Supports NRZ-L, NRZ-I, Manchester, Differential Manchester, and Alternate Mark Inversion (AMI), with matching decoders.
- **Scrambling:** Provides optional B8ZS and HDB3 scrambling and descrambling for AMI signals.
- **Analog-to-digital conversion:** PCM converts samples into fixed-width binary codewords and reconstructs approximate values. DM generates a bitstream using a configurable step size and reconstructs a staircase approximation.
- **Visualization:** Uses Java graphical components to display encoded waveforms and reconstructed analog samples.

3. Design and Assumptions

The application separates data generation, palindrome detection, encoding/decoding, scrambling, modulation, and visualization into dedicated classes. Digital signals are represented as voltage-level sample arrays. PCM uses a configurable number of bits per sample; DM uses a positive step size. Both reconstructed signals are approximations. Input samples are expected to be normalized to the range –1 to 1.

4. How to Run in Visual Studio Code

- Open the project folder in VS Code and ensure a compatible JDK is installed.
- Open the **src** folder in the integrated terminal.
- Compile the project using the command below, then run **java Main**.

```
javac Main.java data\*.java encoding\*.java decoding\*.java scrambling\*.java
modulation\*.java visualization\*.java
```

Use the menus to choose digital or analog input, an encoding/modulation method, and optional scrambling.

5. Testing and Verification

The original report states that integration tests passed for the tested configurations, including PCM with NRZ-L, Delta Modulation, and AMI scrambling with B8ZS and HDB3. Standalone known-case tests for B8ZS and HDB3 also passed. These results apply only to tested inputs and do not prove correctness for every possible input.

6. Reference

- ITT 036, Programming Assignment 1 specification, Autumn 2026.
- Oracle Java Platform Documentation: https://docs.oracle.com/en/java/